Readiness diagnostic: the measurements

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V3 of the readiness diagnostic; V2 is the sixteen-check page, V1 the criteria roll-up

Generated by `scripts/measurement_sheet_build.py` from the framework record.

Each check is one measurement specification: what it asks, the instrument that reads it, what makes it pass, and where the expected behaviour comes from. The fields follow the ISO/IEC/IEEE measurement information model and its older, simpler form, Goal Question Metric.

1. Data offered as a file a program can read (A1)

Field	Value
Question	Is the data offered in a structured, machine-readable format?
What is examined	On the agency's data page, and the files linked from that page that sit on the same website as the page itself.
Instrument	httpx (httpx>=0.27); mode: auto
What the instrument reads	HTTP GET the product page; extract every download link; for each, HEAD and read Content-Type and file extension. Structured iff any of csv/json/parquet/xlsx/xml; PDF-only iff every link is application/pdf.
Evidence kept	response headers + link list + body sha256
Passes when	A link on this data page, on the page's own website, returned a file that the server labeled by its media type, or the address named by its file extension, as a structured data format: CSV, JSON, XML, spreadsheet or Parquet. Every data page that was looked at met this.
Partial when	Some of the agency's data pages linked such a structured file and some did not.
Fails when, and what a fail does not claim	Every linked file on the same website, in the page's HTML as first served (files built by scripts or reached through a form are not seen), was checked, and none was labeled by its media type or its file extension as a structured data format. It is not a claim that the page links no structured file at all (a real data file served with a generic label and no extension is missed), nor that the agency publishes no machine-readable data: data offered on another website, such as a data portal or an API, or reached only through a query tool, is outside this page's same-website links.
Not measured when	The public client could not look at one or more of the linked files, for one of three reasons: the host refused the identified client; the client's own compliance with the site's robots.txt kept it from the file; or it could not confirm the absence because the page carried more same-website links than the cap of twenty-five, or a server or network error occurred.
Expected behaviour comes from	https://www.rfc-editor.org/rfc/rfc9110.txtData on the Web Best Practices (W3C Recommendation, 31 January 2017): "Make data available in a machine-readable, standardized data format that is well suited to its intended or potential use."AI-Ready Data: Ensuring Public Data Meets the Needs of AI and the American Public, The USAFacts Guide to AI-Ready Data for Government Agencies: "Publish datasets in common structured formats (JSON, CSV, machine-readable Excel). Avoid documents (PDFs) that need to be processed and parsed."M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "data in a format that can be easily processed by a computer without human intervention"M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "Agencies must make each of their public data assets machine-readable."
Who else measures this	Google Lighthouse runs this kind of check: anyone can run its automated checks against a named page, its criteria are published, and a result comes back met, not met, or not applicable. It does not run this exact one.
Client	standard user; API user: Nothing: this check needs no key.

2. The whole dataset in one download (A3)

Field	Value
Question	Can the whole dataset be retrieved in one request?
What is examined	On the agency's data page, and the files linked from that page that sit on the same website as the page itself.
Instrument	scrapy (scrapy>=2.11); mode: auto
What the instrument reads	Crawl the product page one hop; classify links as bulk (whole-product archive or full dataset file) vs filtered query.
Evidence kept	link graph + Content-Length per candidate
Passes when	This data page links, on its own website, an archive file (named by its extension) or a file that is not a web page, carries no query in its address, and is at least fifty kibibytes. It is not verified to be the complete product. Every data page that was looked at met this.
Partial when	Some of the agency's data pages linked such a whole-product download and some did not.
Fails when, and what a fail does not claim	Every linked file on the same website, in the page's HTML as first served (a download behind a script-driven button or a form is not seen), was checked, and none is an archive or a non-web-page file without a query of at least fifty kibibytes. The two tests are stand-ins: a link with a query may still deliver the whole product, and a complete product smaller than fifty kibibytes fails the size floor. It is not a claim that the agency's downloads are partial, nor that no bulk download exists elsewhere, such as a data portal, an FTP site, or a bulk endpoint on a data interface.
Not measured when	The public client could not look at one or more of the linked files, for one of three reasons: the host refused the identified client; the client's own compliance with the site's robots.txt kept it from the file; or it could not confirm the absence because the page carried more same-website links than the cap of twenty-five, or a server or network error occurred.
Expected behaviour comes from	Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Enable consumers to retrieve the full dataset with a single request."OMB M-23-22: Delivering a Digital-First Public Experience: "making that information available as machine-readable data that can be accessed in bulk"OMB M-23-22: Delivering a Digital-First Public Experience: "If an agency detects significant public interest in scraping information from web pages, the agency should strongly consider making that information available as machine-readable data that can be accessed in bulk"
Who else measures this	Google Lighthouse runs this kind of check: anyone can run its automated checks against a named page, its criteria are published, and a result comes back met, not met, or not applicable. It does not run this exact one.
Client	standard user; API user: Nothing: this check needs no key.

3. The data page is in the site's list of pages (A5)

Field	Value
Question	Is the data page listed in the host's sitemap?
What is examined	The sitemap files at the standard locations on the agency's own site (a sitemap, a sitemap index, and any sitemap the site's robots.txt names on that same site), plus an llms.txt and the well-known discovery paths.
Instrument	ultimate-sitemap-parser (ultimate-sitemap-parser>=1.0); mode: auto
What the instrument reads	HTTP GET /sitemap.xml (and the path robots.txt declares), /llms.txt, /.well-known/; parse the sitemap and test whether the product URL is covered.
Evidence kept	sitemap tree + coverage verdict for the product URL
Passes when	Every data page looked at was found listed: a sitemap served on the agency's own site carries the page's address exactly as the diagnostic records it.

Field	Value
Partial when	Some of the agency's data pages are listed in a served sitemap and some are not.
Fails when, and what a fail does not claim	For every data page looked at, the sitemaps served on the agency's own site were read and none lists the page's address in exactly the recorded form (a trailing slash, redirect target or query-string variant does not match), or the only discovery file served is an llms.txt (which by its proposal summarises a site rather than listing every page), or no sitemap is served at the standard locations; the reason names which. It is not a claim that the page is absent from the sitemap in every form, nor that search engines cannot find the page, because well-linked pages are found without a sitemap.
Not measured when	The public client could not read one or more of the discovery files, for one of three reasons: the host refused the identified client, the client's own robots.txt compliance kept it from reading the file, or it could not confirm an absence or hit a server or network error.
Expected behaviour comes from	https://developers.google.com/search/docs/crawling-indexing/sitemaps/overviewOMB M-23-22: Delivering a Digital-First Public Experience: "Agencies should employ best practices to improve crawling or indexing of web content, including using sitemaps, robots.txt files"Sitemaps XML format (sitemaps.org protocol): "The location of a Sitemap file determines the set of URLs that can be included in that Sitemap."https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview: "You might not need a sitemap if"https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview: "in most cases, your site will benefit from having a sitemap"Sitemaps XML format (sitemaps.org protocol): "You can specify the location of the Sitemap using a robots.txt file."Sitemaps XML format (sitemaps.org protocol): "a Sitemap is expected to have URLs from a single host only"https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview: "You might not need a sitemap if: Your site is "small". By small, we mean about 500 pages or fewer on your site."https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview: "Your site is comprehensively linked internally."https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview: "Your site is large. Generally, on large sites it's more difficult to make sure that every page is linked by at least one other page on the site."https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview: "A sitemap tells search engines which pages and files you think are important in your site"
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

4. A published description of the agency's data service (A2)

Field	Value
Question	Does the agency publish a machine-readable description of its data API?
What is examined	At the data-interface address recorded for the agency in the project's records (read from the agency's own pages or seeded from a named source such as a developer page or a public catalog), and any description document recorded beside that address.
Instrument	httpx (httpx>=0.27); mode: auto
What the instrument reads	HTTP GET the documented API base and any OpenAPI/Swagger document; record auth scheme, declared rate limits, and whether the description is machine-readable.

Field	Value
Evidence kept	OpenAPI document + response headers
Passes when	A machine-readable OpenAPI or Swagger description document is served at the data-interface address recorded for this agency, and the result names that address. Every address that was looked at met this.
Partial when	Some of the recorded addresses served such a description and some did not.
Fails when, and what a fail does not claim	No OpenAPI or Swagger document was found at the data-interface addresses recorded for this agency; where no address is recorded at all, the fail says only that none was found in the sources searched. It is not a claim that the agency has no data interface, that its interface is undocumented, that it publishes no machine-readable description anywhere (a description at a developer page, or in another format such as a catalog document at the interface's root, is not looked for), or that its interface refuses programs.
Not measured when	The public client could not look at one or more recorded addresses, for one of three reasons: the host refused the identified client; the client's own compliance with the site's robots.txt kept it from the recorded address; or it could not confirm an absence because a recorded address was left unverified or a seed source unsearched, or a server or network error occurred.
Expected behaviour comes from	Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Provide complete information on the Web about your API."Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "It will also enables machines to access the API documentation in order to help developers build API client software."OpenAPI Specification (latest, spec.openapis.org): "allows both humans and computers to discover and understand the capabilities of a service"RFC 9727: api-catalog: A Well-Known URI and Link Relation to Help Discovery of APIs: "they SHOULD make that metadata available at the API endpoint URIs they have listed"Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Check that every call enabled by your API is described in your documentation. Make sure you provide details of what parameters are required or optional and what each call returns."OpenAPI Specification (latest, spec.openapis.org): "This string MUST be the version number of the OpenAPI Specification that the OpenAPI document uses."RFC 9727: api-catalog: A Well-Known URI and Link Relation to Help Discovery of APIs: "It is RECOMMENDED that the API catalog also includes useful metadata, such as usage policies, API version information, links to the OpenAPI Specification [OAS] definitions for each API"
Who else measures this	Google Lighthouse runs this kind of check: anyone can run its automated checks against a named page, its criteria are published, and a result comes back met, not met, or not applicable. It does not run this exact one.
Client	standard user; API user: The description document is usually public and needs no key; but this check reads the recorded data-interface address, which could itself require a key, so registering as a member of the public and presenting one could let the client reach a document a bare request cannot.

5. Whether the site's visitor rules let AI tools in (A4)

Field	Value
Question	Does the host's declared crawler policy let SI systems' crawlers fetch the data page?

Field	Value
What is examined	The single robots.txt file at the root of the agency's website, taken one host per agency.
Instrument	protego (protego>=0.3); mode: auto
What the instrument reads	HTTP GET /robots.txt; parse; evaluate Allow/Disallow for the product path per user-agent in the AI-crawler list (GPTBot, ClaudeBot, PerplexityBot, Google-Extended, CCBot, Bingbot).
Evidence kept	robots.txt body + sha256 + per-UA verdict
Passes when	The host's robots.txt, as stated, allows each of eight listed crawler tokens to fetch the data page's path. Every data page that was looked at met this.
Partial when	For some of the agency's data pages the stated policy allowed all eight listed crawler tokens and for some it did not.
Fails when, and what a fail does not claim	Either the host serves no robots.txt, which under the robots.txt standard actually permits every crawler, or its robots.txt tells at least one of the listed crawler tokens not to fetch the path. Neither is a claim that crawlers cannot reach the data: robots.txt is a stated policy, not a barrier, and telling a training-only crawler not to fetch does not keep a page out of answer engines.
Not measured when	The public client could not read the robots.txt, for one of three reasons: the host refused the identified client (three of sixteen hosts on the cycle of record); the client's own compliance with the site's robots.txt kept it from the file; or it could not confirm the file's state because of a server or network error.
Expected behaviour comes from	RFC 9309: Robots Exclusion Protocol: "then the crawler MAY access any resources on the server."RFC 9309: Robots Exclusion Protocol: "These rules are not a form of access authorization."Google Search Central: AI features and your website: "is the control for site owners to manage access to how their sites are crawled for Search"Google Search Central: AI features and your website: "To limit AI training and grounding in some of Google's other systems, read more about Google-Extended"OpenAI: Overview of OpenAI crawlers (developers.openai.com/api/docs/bots): "Disallowing GPTBot indicates a site's content should not be used in training generative AI foundation models."OpenAI: Overview of OpenAI crawlers (developers.openai.com/api/docs/bots): "To help ensure your site appears in search results, we recommend allowing OAI-SearchBot in your site's robots.txt file"Does Anthropic crawl data from the web, and how can site owners block the crawler? (Anthropic support): "it signals that the site's future materials should be excluded from our AI model training"RFC 9309: Robots Exclusion Protocol: "Crawlers MUST use case-insensitive matching to find the group that matches the product token and then obey the rules of the group."RFC 9309: Robots Exclusion Protocol: "For example, in the context of HTTP, such status codes are in the 400-499 range."Google Search Central: AI features and your website: "AI is built into Search and integral to how Search functions, which is why robots.txt directives for Googlebot is the control for site owners to manage access to how their sites are crawled for Search."
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

6. The page labels itself as a dataset (A6)

Field	Value
Question	Does the data page carry structured markup that tells a machine it describes a dataset?

Field	Value
What is examined	The data page itself, reading the structured markup embedded in the page as it is served.
Instrument	extruct + pyshacl (extruct>=0.17); mode: auto
What the instrument reads	Extract embedded JSON-LD / microdata / RDFa from the product page; test for schema.org Dataset or DCAT; validate the DCAT graph against DCAT-AP SHACL shapes.
Evidence kept	extracted graph + SHACL report
Passes when	Every data page looked at carried markup, as served before any script runs, that declares a dataset with a name and a description (and any distribution it declares carries an access address).
Partial when	Some of the agency's data pages carry conforming dataset markup and some do not.
Fails when, and what a fail does not claim	For every data page looked at, the markup as served before any script runs does not declare a dataset, or declares one that fails the minimal shapes (a name or description missing, or a declared distribution without an access address); the reason names the violation. It is not a claim that a dataset declared with a name and description is missing when the failing shape is the distribution's, nor that the agency's datasets are absent from dataset search, because catalog pages, data.gov records and other pages may carry the markup.
Not measured when	The public client could not read one or more pages, for one of three reasons: the host refused the identified client, the client's own robots.txt compliance kept it from reading the page, or the markup was declared but its graph could not be loaded to confirm the result, a server or network error.
Expected behaviour comes from	https://developers.google.com/search/docs/crawling-indexing/javascript/javascript-seo-basicsGoogle Search Central: Dataset (Dataset, DataCatalog, DataDownload) structured data: "We recommend that you add structured data about a dataset to the canonical pages."Google Search Central: Dataset (Dataset, DataCatalog, DataDownload) structured data: "or equivalent structures represented in W3C's Data Catalog Vocabulary (DCAT) format"Google Search Central: Dataset (Dataset, DataCatalog, DataDownload) structured data: "You must include the required properties for your content to be eligible for display as a rich result."Google Search Central: Dataset (Dataset, DataCatalog, DataDownload) structured data: "A short summary describing a dataset."Google Search Central: Dataset (Dataset, DataCatalog, DataDownload) structured data: "A descriptive name of a dataset."Google Search Central: Dataset (Dataset, DataCatalog, DataDownload) structured data: "Required properties, description / A short summary describing a dataset."Google Search Central: Dataset (Dataset, DataCatalog, DataDownload) structured data: "The same schema.org vocabulary can also be used in RDFa 1.1 or Microdata syntaxes."schema.org: Dataset: "A body of structured information describing some topic(s) of interest."
Who else measures this	Google Lighthouse is the nearest precedent: anyone can run its automated page-level checks against a named page and get back met, not met, or not applicable against published criteria.
Client	standard user; API user: Nothing: this check needs no key.

7. The release date, and a link to the latest release (A8)

Field	Value
Question	Can a machine tell which release of the data it is reading, and find the latest one?
What is examined	The data page itself (its structured date markup and the server's last-modified header) and up to three links on the same site that point to the latest version.
Instrument	httpx + extruct (httpx>=0.27); mode: auto
What the instrument reads	Read Last-Modified and any dcterms:modified / schema.org dateModified in the markup; test whether a latest-vintage pointer resolves.
Evidence kept	headers + markup date fields
Passes when	Every data page looked at carried a release or modification date in its markup (in the dateModified or datePublished field, or in temporalCoverage, a case this diagnostic marks as its own choice), and a latest-version link on the same site resolved.
Partial when	Some of the agency's data pages are dated in markup with a resolving same-site latest-version link and some are not.
Fails when, and what a fail does not claim	For every data page looked at, the markup does not carry a date in the three fields read (dateModified, datePublished, temporalCoverage), or carries one but none of up to three recognised same-site latest-version links resolves. It is not a claim that the agency does not date or version its releases: a version number, a date in another markup field, in visible text, a PDF or a catalog record, and a latest link worded otherwise, beyond the third, or on another site are not read.
Not measured when	The public client could not read one or more pages, for one of three reasons: the host refused the identified client, the client's own robots.txt compliance kept it from reading the page, or a latest-version link could not be confirmed, a server or network error.
Expected behaviour comes from	https://www.rfc-editor.org/rfc/rfc9110.txtData on the Web Best Practices (W3C Recommendation, 31 January 2017): "Assign and indicate a version number or date for each dataset."Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Check if a computer application can automatically detect/discover and process the unique version number or date of a dataset or distribution."Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Humans and software agents will be able to refer to specific versions of a dataset and to concepts such as a 'dataset series' and 'the latest version'."https://www.rfc-editor.org/rfc/rfc9110.txt: "provides a timestamp indicating the date and time at which the origin server believes the selected representation was last modified"DCAT-US Schema v1.1 (Project Open Data Metadata Schema) - resources.data.gov: "Most recent date on which the dataset was changed, updated or modified."Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Assign URIs to individual versions of datasets as well as to the overall series."
Who else measures this	Google Lighthouse is the nearest precedent: anyone can run its automated page-level checks against a named page and get back met, not met, or not applicable against published criteria.
Client	standard user; API user: Nothing: this check needs no key.

8. A front door built for programs (A9)

Field	Value
Question	Does the agency serve an entry point built for machine clients (an API description, an agent-protocol endpoint, an llms.txt) where such clients look first?

Field	Value
What is examined	A fixed handful of machine addresses this project checks on the agency's site: the recorded API base and its description, the /.well-known/mcp path, and /llms.txt.
Instrument	httpx (httpx>=0.27); mode: auto
What the instrument reads	Probe for a machine-first entry point: OpenAPI at the documented base, /.well-known/mcp or an advertised MCP/A2A endpoint, /llms.txt.
Evidence kept	probe matrix + any served descriptor
Passes when	For every data page looked at, at least one of the addresses this project checks returned content that is not an HTML page.
Partial when	For some of the agency's data pages at least one checked address returns non-HTML content and for some none does.
Fails when, and what a fail does not claim	For every data page looked at, none of the handful of addresses this project checks, by a plain request with no key and no accept header, returned non-HTML content. The recorded API address may be stale or wrong, and an API that answers a bare request with an HTML page or an error page fails the same way; it is not a claim that the agency has no API, that its API is down, or that it lacks machine access.
Not measured when	The public client could not read one or more addresses, for one of three reasons: the host refused the identified client, the client's own robots.txt compliance kept it from reading, or the recorded API base could not be reached to confirm an absence, a server or network error.
Expected behaviour comes from	not specified
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: This check requests with no key, so a registered public client that presents a key could change the result here: an API that answers only a key-bearing request could return machine content on a second look where the bare request got an HTML or error page.

9. The interactive data tool answers honestly (A10)

Field	Value
Question	Does the agency's interactive data tool answer a machine client honestly: real content at a deep link, and a real error for a route that does not exist?
What is examined	At the one interactive data tool the agency offers, looked at from the site level: its direct link to a product, and a made-up address beneath that tool.
Instrument	lighthouse (lighthouse-cli>=12); mode: auto
What the instrument reads	For an interactive data tool: request a deep link and an invalid route; compare status codes and pre-JS HTML; a soft-404 is HTTP 200 with an error shell. Render once to compare pre- and post-JS DOM.
Evidence kept	status codes + raw HTML + rendered DOM
Passes when	The direct link came back as a working page carrying at least two thousand characters of text before any script ran, and the made-up address correctly came back as an error rather than a normal "OK" page.
Partial when	Some of the agency's data pages that were looked at met this and some did not.
Fails when, and what a fail does not claim	One of three things, named in the result: a made-up address came back as a normal "OK" page instead of an error (which may be a platform or vendor default rather than the agency's choice); the direct link itself came back as an error; or fewer than two thousand characters of text were present before scripts ran. It is not a claim that the tool is unusable in a browser or by a crawler that runs JavaScript, nor a finding about the agency's intent.
Not measured when	The public client could not look at one or more pages, for one of three reasons: the host refused the identified client; the client's own robots.txt

Field	Value
	compliance kept it from reading the page; or it could not confirm an absence, or a server or network error stopped it.
Expected behaviour comes from	https://www.rfc-editor.org/rfc/rfc9110.txthttps://developers.google.com/crawling/docs/troubleshooting/http-status-codeshttps://developers.google.com/search/docs/crawling-indexing/javascript/javascript-seo-basicshttps://www.rfc-editor.org/rfc/rfc9110.txt: "indicates that the origin server did not find a current representation for the target resource"https://developers.google.com/crawling/docs/troubleshooting/http-status-codes: "Search Console will show a soft 404 error"https://developers.google.com/search/docs/crawling-indexing/javascript/javascript-seo-basics: "Googlebot queues all pages with a 200 HTTP status code for rendering"
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

10. What the site states about automated access (A11-declared)

Field	Value
Question	What does the host declare about crawler access to the data page, in both places a declaration can live (robots.txt and the page's robots meta tag)?
What is examined	At the site's robots.txt file and at the robots meta tags carried on the data page.
Instrument	protego (protego>=0.3); mode: auto
What the instrument reads	The DECLARED layer only: robots.txt and meta-robots directives for the product path. The enforced and observed layers need edge/WAF logs and are agency_instrumented.
Evidence kept	robots.txt + meta-robots + per-UA declared verdict
Passes when	This records what the agency declares; it is read but not scored. What it records as permitting: both the robots.txt and the page's meta tags, as written, allow each of the eight listed crawler names.
Partial when	This is a declaration that is read but not scored, so no partial result is given.
Fails when, and what a fail does not claim	It records one of three things, named in the result: the robots.txt disallows a listed name (for five of the eight names that is a request not to use content for training, not a block on reading); the page's meta tags restrict it; or no robots.txt is served at all, which declares no restriction and under the standard permits every crawler. A fail is therefore not a finding that the agency declared a policy restricting crawler access, and it is about declared policy only, not about whether any crawler is actually served or refused.
Not measured when	The declared layer could not be looked at, for one of three reasons: the host refused the identified client; the client's own robots.txt compliance kept it from reading the page; or it could not confirm an absence, or a server or network error stopped it.
Expected behaviour comes from	https://developers.google.com/search/docs/crawling-indexing/robots-meta-tagRFC 9309: Robots Exclusion Protocol: "then the crawler MAY access any resources on the server."https://developers.google.com/search/docs/crawling-indexing/robots-meta-tag: "rule applies to search engine crawlers"OpenAI: Overview of OpenAI crawlers (developers.openai.com/api/docs/bots): "Disallowing GPTBot indicates a site's content should not be used in training generative AI foundation models."RFC 9309: Robots Exclusion Protocol: "These rules are not a form of access authorization."https://developers.google.com/search/docs/crawling-indexing/robots-meta-tag: "these settings can be

Field	Value
	read and followed only if crawlers are allowed to access the pages that include these settings"OpenAI: Overview of OpenAI crawlers (developers.openai.com/api/docs/bots): "To help ensure your site appears in search results, we recommend allowing OAI-SearchBot in your site's robots.txt file"Does Anthropic crawl data from the web, and how can site owners block the crawler? (Anthropic support): "it signals that the site's future materials should be excluded from our AI model training"Cloudflare AI Crawl Control: Manage AI crawlers: "For each AI crawler, you can choose to allow or block access."
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

11. The site serves what its own rules permit (A12)

Field	Value
Question	Does the host enforce what it declares?
What is examined	At the site's robots.txt file and at the data page's web address, both requested as the same identified client.
Instrument	http + robots (httpx>=0.27 + protego (scripts/scan_preflight.py)); mode: auto
What the instrument reads	Per host: GET /robots.txt and parse it; then GET the product path under the identified UA. Coherent iff robots.txt permits the path for this UA AND the host serves it. Incoherent iff robots.txt permits and the host answers a status in `manners.unobservable_statuses` (401/403/407/429). A robots.txt that DISALLOWS the path is not incoherence, it is A4's measurement, and this indicator is `not_applicable` there.
Evidence kept	robots.txt body + per-UA verdict + the response status for the same path
Passes when	This check is under review and is not scored yet. What it would record as a pass: the robots.txt permits this client on the data page's path and the host actually served the page to it.
Partial when	This check is under review and is not scored yet, so no result is given.
Fails when, and what a fail does not claim	It would record one of three things, named in the result: the host answered a refusal to this client's request for the robots.txt file; the robots.txt permits the client on the path and the host answered a refusal for that same path; or no robots.txt is served, which under the standard declares permission and involves no contradiction (the page may well have been served). Only the second is a host doing other than it declared; a rate-limit refusal may be temporary rather than a policy; a temporary error or server failure is never a fail here; and it is about one identified client, not every crawler.
Not measured when	The public client could not settle the page, for one of three reasons: the host refused the identified client; the client's own robots.txt compliance kept it from reading; or it could not confirm an absence, or a server or network error stopped it, here, no response to either request or a server error on the page request.
Expected behaviour comes from	RFC 9309: Robots Exclusion Protocol: "These rules are not a form of access authorization."RFC 9309: Robots Exclusion Protocol: "The Robots Exclusion Protocol is not a substitute for valid content security measures."RFC 9309: Robots Exclusion Protocol: "then the crawler MAY access any resources on the server."RFC 9309: Robots Exclusion Protocol: ""Unavailable" means the crawler tries to fetch the robots.txt file and the server responds with status codes indicating that the resource in question is unavailable."RFC 9309: Robots Exclusion Protocol: "For example, in the context of HTTP, such status codes are in the 400-499 range."

Field	Value
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

12. From a data page to the data service, terms, change log and inventory (A13)

Field	Value
Question	Can a machine client that starts at a data page, and knows nothing else, reach the four objects an agency publishes for machines (its API, the API's terms, its changelog, its data inventory)?
What is examined	At one of the agency's data pages, its links and its response headers, and at the recorded locations where each of the four objects was reported to sit.
Instrument	none of its own (`params.link_probe.legs_served`) (the shared link probe's product-page Observation and the A2, D1, F4 and D4 Observations (`RULE-A13-v1.CONSUMES`)); mode: auto
What the instrument reads	Per surface, per object (the API, its terms, its changelog, the inventory): pass when an object that exists (an Observation at a recorded location served it, or the record says verified) has a recorded URL among the product page's links, on-host or off-host and past the link probe's cap, or sits at a convention location (the well-known api-catalog URI served as a Linkset, or a `Link: rel="api-catalog"` header, RFC 9727; the data.json inventory at the host root, OMB M-13-13); fail when it exists, the page was read whole and neither holds; error when existence is not settled, the page was not read, or the API's convention location was not fetched; not_applicable when no object is shown to exist. The surface's verdict is the first of fail, error, pass, not_applicable that any object returns.
Evidence kept	the product page's link set and response headers, plus the recorded locations and the fetches of them
Passes when	Every object the agency is shown to publish is either linked from this data page or sits at its published standard address.
Partial when	Some of the agency's data pages that were looked at reached every published object and some did not.
Fails when, and what a fail does not claim	At least one object the agency is shown to publish (named in the result) is neither linked from this data page nor at a standard address the check looked at, the inventory at /data.json on the data page's own host (an inventory on the main www site while the page sits on a subdomain is not matched there), or the API catalog at /.well-known/api-catalog. For the service's terms and the changelog no published standard exists, so a fail on either rests only on this project's "must be linked" rule and breaks no published convention. It is not a claim that the object is missing, nor that a client told where to look cannot use it.
Not measured when	The public client could not settle one or more objects, for one of three reasons: the host refused the identified client; the client's own robots.txt compliance kept it from reading; or it could not confirm an absence, or a server or network error stopped it, here, an object whose existence was not settled, a page not served or its links not read, or an API catalog address no collector has fetched yet.
Expected behaviour comes from	RFC 9727: api-catalog: A Well-Known URI and Link Relation to Help Discovery of APIs: "SHALL resolve an HTTPS GET request to /.well-known/api-catalog and return an API catalog document"RFC 9727: api-catalog: A Well-Known URI and Link Relation to Help Discovery of APIs: "This identifies a target resource that represents a list of APIs available from the Publisher of the link context."M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management:

Field	Value
	"each agency must post the inventory in JSON format, or a successor format, to www.[agency].gov/data.json"RFC 9727: api-catalog: A Well-Known URI and Link Relation to Help Discovery of APIs: "the Publisher "example" could include the api-catalog link relation in the HTTP header and/or content payload"
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

13. The methods are readable as plain text (B3)

Field	Value
Question	Can a machine reach the methodology behind the data and read it?
What is examined	The data product page itself, and any link on it whose web address or wording contains the word "methodology" (up to twenty-five such links).
Instrument	httpx (httpx>=0.27); mode: auto
What the instrument reads	Fetch the methodology document; test whether it is structured text (HTML/markdown) or PDF-only, and whether it is retrievable without JS.
Evidence kept	Content-Type + body sha256
Passes when	Every data page that was looked at links to a methodology document served as a web page or plain text, readable before any script runs, with enough text and not mostly links.
Partial when	Some of the agency's data pages that were looked at met this and some did not.
Fails when, and what a fail does not claim	For every data page looked at, no methodology link (up to twenty-five followed) led to a document meeting the project's text measures: at least two thousand visible characters before any script runs and no more than a quarter of the text sitting in links. A short but readable page, or a chapter hub, can fall short of those measures. It is not a claim that the methodology needs scripts or cannot be read, nor that the agency publishes no methodology: one linked under another name, such as "technical documentation" or "about the data", is simply not found.
Not measured when	The public client could not finish looking at one or more pages, for one of three reasons: the host turned the identified client away; the client's own honoring of the site's robots.txt stopped it reading the page; or a server or network error, or methodology links it could not follow, left the result unconfirmed.
Expected behaviour comes from	AI-Ready Data: Ensuring Public Data Meets the Needs of AI and the American Public, The USAFacts Guide to AI-Ready Data for Government Agencies: "Ensure text documentation (data dictionaries, methodology) are accessible to both humans and AI/web crawlers."AI-Ready Data: Ensuring Public Data Meets the Needs of AI and the American Public, The USAFacts Guide to AI-Ready Data for Government Agencies: "AI models need context, such as variable definitions, collection methodology, and version history"OMB M-23-22: Delivering a Digital-First Public Experience: "An agency should develop online content in a non-HTML format only if necessitated by a specific user need."AI-Ready Data: Ensuring Public Data Meets the Needs of AI and the American Public, The USAFacts Guide to AI-Ready Data for Government Agencies: "Data Must Be Machine Understandable with Easily Retrievable"AI-Ready Data: Ensuring Public Data Meets the Needs of AI and the American Public, The USAFacts Guide to AI-Ready Data for Government Agencies: "Disclose methodologies for derived or transformed datasets, ensuring clarity on how figures are generated."
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

14. The product is listed in the agency's data inventory (D4)

Field	Value
Question	Is the data product listed in the agency's public data inventory?
What is examined	The agency's public data inventory at each location on record for it, its own data.json file, or a catalog recorded for the agency.
Instrument	project-open-data-validator (json-schema (Project Open Data v1.1 schema)); mode: auto
What the instrument reads	Fetch /data.json (Project Open Data catalog); validate against the POD v1.1 schema; test whether the product appears in it.
Evidence kept	catalog document + validation report + membership verdict
Passes when	Every data page that was looked at appears in a record that names its address in a DCAT-US address field, inside an inventory that conforms to the federal metadata schema.
Partial when	Some of the agency's data pages that were looked at met this and some did not.
Fails when, and what a fail does not claim	For every data page looked at, one of three things named in the result's reason: the recorded inventory was read and no record names this page's address; or the page is listed but the inventory as a whole fails the schema, possibly in an unrelated record; or no inventory is served at any recorded address. It is not a claim that the dataset is missing from the inventory under some other address, nor that the agency has no inventory at an address not on record for it, such as a parent department's.
Not measured when	The public client could not finish looking at one or more pages, for one of three reasons: the host turned the identified client away; the client's own honoring of the site's robots.txt stopped it reading the inventory; or a recorded inventory could not be confirmed, or a starting source was not searched, because of a server or network error.
Expected behaviour comes from	M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "develop and maintain a comprehensive data inventory that accounts for all data assets"M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "must employ the metadata schema approved by OMB and available on resources.data.gov"M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "including in the agency's comprehensive data inventory the URL or persistent identifier"M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "Each agency is required to develop and maintain a comprehensive data inventory of all data assets maintained by the agency."M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "commonly known as the JSON format"M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "host it publicly on the agency's website at the address: www.[agency].gov/data.json"M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "When an agency chooses to disseminate a public data asset through the Federal Data Catalog, they do so by including in the agency's comprehensive data inventory the URL or persistent identifier"DCAT-US Schema v1.1 (Project Open Data Metadata Schema) - resources.data.gov: "This field is not intended for an agency's homepage (e.g. www.agency.gov), but rather if a dataset has a human-friendly"
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.

15. The terms of reuse, under a recognised name (D1)

Field	Value
Question	Can a machine tell on what terms it may use the data?
What is examined	The licence field in the data page's markup, a licence link sent in the page's response headers, and, when an interface for developers is on record, that interface's terms page.
Instrument	extruct + httpx (extruct>=0.17); mode: auto
What the instrument reads	Read the licence from schema.org/DCAT markup, an HTTP Link header, and the API's terms endpoint; test whether it is a recognised identifier (SPDX or a known URL) rather than free text.
Evidence kept	licence field + resolution verdict
Passes when	Every data page that was looked at names a licence by a recognised identifier in at least one of those places.
Partial when	Some of the agency's data pages that were looked at met this and some did not.
Fails when, and what a fail does not claim	For every data page looked at, none of the places read names a licence by a recognised identifier: the page's markup, its licence link header, and the developer-interface terms page only when a terms address is on record for the agency (when none is on record, that place was not read). It is not a claim that the data is unlicensed or restricted: federal data is generally public domain, and the agency's own data inventory, where federal guidance says the status may be recorded, is not read by this check.
Not measured when	The public client could not finish looking at one or more pages, for one of three reasons: the host turned the identified client away; the client's own honoring of the site's robots.txt stopped it reading the page; or the developer-interface terms address was not yet settled, or a server or network error left the result unconfirmed.
Expected behaviour comes from	Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Check if a user agent can automatically detect or discover the data license of the dataset."M-25-05: Phase 2 Implementation of the Evidence Act, Open Government Data Access and Management: "Agencies may satisfy this requirement by indicating, in the comprehensive data inventory, that the data asset is in the public domain"DCAT-US Schema v1.1 (Project Open Data Metadata Schema) - resources.data.gov: "The license or non-license (i.e. Public Domain) status with which the dataset or API has been published."Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Provide a link to or copy of the license agreement that controls use of the data."
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: A public key could let the client reach a developer interface's terms page that is shown only to registered developers; the page-markup and response-header checks need no key.

16. A change history a program can follow (F4)

Field	Value
Question	Can a machine learn what changed between releases?
What is examined	Each change-log or release-notes location on record for the agency.
Instrument	httpx (httpx>=0.27); mode: auto
What the instrument reads	Fetch any changelog or release-notes endpoint; test whether it is machine-readable and carries a revision class per entry.
Evidence kept	changelog document + parse verdict
Passes when	Every data page that was looked at has a change history served as a JSON, Atom or RSS feed, with at least half its entries saying what kind of change they record.

Field	Value
Partial when	Some of the agency's data pages that were looked at met this and some did not.
Fails when, and what a fail does not claim	For every data page looked at, no such feed was found at the recorded locations: there is no change history there, or there is one published as an ordinary web page, or its entries do not say what kind of change they are, or the recorded locations were missing or wrong. The project's format and labelling floors decide two of these, so a precise web-page change log that a machine can read, and that a Data on the Web Best Practice accepts, still fails here. A fail is not a claim that the version history is absent, incomplete or unreadable, nor that the agency does not publish revisions.
Not measured when	The public client could not finish looking at one or more pages, for one of three reasons: the host turned the identified client away; the client's own honoring of the site's robots.txt stopped it reading the page; or a starting source was not searched, or a server or network error left the result unconfirmed.
Expected behaviour comes from	Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Provide a complete version history that explains the changes made in each version."Data on the Web Best Practices (W3C Recommendation, 31 January 2017): "Check that a list of published versions is available as well as a change log describing precisely how each version differs from the previous one."AI-Ready Data: Ensuring Public Data Meets the Needs of AI and the American Public, The USAFacts Guide to AI-Ready Data for Government Agencies: "Implement versioning so AI developers can track updates over time."
Who else measures this	No public tool runs this exact check.
Client	standard user; API user: Nothing: this check needs no key.